

1(a)	A – G protein ; B – cyclic AMP / cAMP ; C – aquaporin ; I water channel	3
1(b)	any three from: 1 dehydration / not drinking enough water ; 2 ingesting salty food / description ; 3 more, ions / glucose / amino acids / solute, (in blood) ; 4 sweating / perspiration / evaporation (of water) ; 5 AVP ; e.g. disease where ADH is not, made / secreted	3
1(c)	any two from: 1 increase in / large, urine volume or more frequent urination or dilute / less concentrated, urine ; 2 fatigue ; 3 feelings of thirst / dry mouth ; 4 AVP ; e.g. dehydration change in blood pressure	2

Question	Answer	Marks
1(d)	any two from: 1 few / no, aquaporins added to membrane / P ; 2 reduced / low, permeability of, membrane / P, to water ; R impermeable to water 3 less water moves through, membrane / P ; ecf for mp1 if C named as water channels in 1(a)	2
1(e)	(gene is on X chromosome) X (chromosome with recessive allele) inherited from mother ; Y (chromosome) inherited from father ;	2